MoonWalker Weekly Meeting

9/25 - 9/30 Summary

Elijah, Rowel, Martin

Progress Update

- Elijah
  - Performance testing for Isaac Sim
    - Ran sample tests and HSML-Models available in GitHub
  - Research on Tailscale and its use case for the project
  - Worked on diagrams and workflows
- Martin
  - Picked up Dr. Li's laptop (9/30/25) and setting up Unity
    - May need an admin account so I'll be visiting the IT office again
  - Setting up HSML API locally on the new laptop and connect w/ Unity
  - Testing MySql DB setup locally as well
- Rowel
  - Set up Digital Twin server for testing (Installed WSL2 with Ubuntu, Docker, Java, Kafka)
  - Ran Kafka on the Digital Twin Server (Created Kafka topics, Sent messages to Producers verified them on Consumer side)

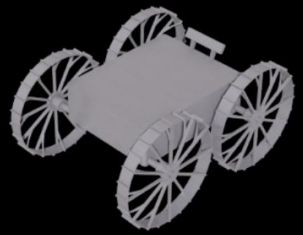
Isaac Sim Full 5.0.0 - D:/School/Research/Current Research/NASA-JPL Digital Twin Interopability/Omniverse Isaac Sim/4WD_Contrl.usd*

File Edit Create Window Tools Utilities Layouts Help

LIVE CACHE: NEW VERSION DETECTED

RTX - Real-Time | Perspective | Default

FPS: 115.84, Frame time: 8.63 ms
NVIDIA GeForce RTX 4070 Laptop GPU: 1.9 GiB used, 5.1 GiB available
Process Memory: 5.5 GiB used, 3.6 GiB available
1280x720



Y Z X
cm

Stage | Layer | Render Settings

Search

Name (Old to New)	Type
World (defaultPrim)	Xform
CADRE_4WD_Controllable	Xform
NewCADRE	Xform
CADRE_Textured_1	Xform
material	Scope
CADRE_Wheel_FR_1	Xform
CADRE_Wheel_BR_1	Xform
CADRE_Wheel_BR	Xform
Body1	Xform
Body1	Mesh
CADRE_Wheel_FL_1	Xform
CADRE_Wheel_BL_1	Xform
CADRE_Wheel_BL	Xform
Body1	Xform
Body1	Mesh
CADRE_Chasis_1	Xform
CADRE_Chasis	Xform
Body1	Xform
Body1	Mesh
ActionGraph	OmniGraph

Property

Search

Console | Content

Import | D:/School/Research/Current Research/NASA-JPL Digital Twin Interopability/Omniverse Isaac Sim/ | Search

- kafka_2.13-4.1.0
- Omniverse Isaac Sim
  - isaac sim
  - Setup Documenta
- Papers
- RAGGY
- Past Research
- Research Paper To Read
- Resume
- SteamLibrary
- Workouts

isaac sim | Setup Docu...ation | 4WD_Contrl.usd | CADREDemo.usd | compatibility-check.png | drive-downl | isaac-sim-co | isaac-sim-st | Sh

Omniverse Isaac Sim

Date Modified	09/24/2025 02:26PM
Created by	
Modified by	
File size	4.00 KB

File Information

No Additional Information

Checkpoints

82°F Sunny | Search | 2:37 PM 9/24/2025

Creating Kafka Topics in Ubuntu / Sending Messages to Producer

```
Windows PowerShell   Ubuntu   Ubuntu
moonwalker@DESKTOP-TOD84J7:~$ bin/kafka-topics.sh --create --topic quickstart-events --bootstrap-server loca
lhost:9092
-bash: bin/kafka-topics.sh: No such file or directory
moonwalker@DESKTOP-TOD84J7:~$ cd mnt/c
-bash: cd: mnt/c: No such file or directory
moonwalker@DESKTOP-TOD84J7:~$ cd /mnt/c
moonwalker@DESKTOP-TOD84J7:/mnt/c$ cd ~
moonwalker@DESKTOP-TOD84J7:~$ cd kafka_2.1.3-4.1.0
-bash: cd: kafka_2.1.3-4.1.0: No such file or directory
moonwalker@DESKTOP-TOD84J7:~$ cd kafka_2.13-4.1.0
moonwalker@DESKTOP-TOD84J7:~/kafka_2.13-4.1.0$ bin/kafka-topics.sh --create --topic quickstart-events --boot
strap-server localhost:9092
Created topic quickstart-events.
moonwalker@DESKTOP-TOD84J7:~/kafka_2.13-4.1.0$ bin/kafka-topics.sh --describe --topic quickstart-events --bo
otstrap-server localhost:9092
Topic: quickstart-events        TopicId: GEbWCSR9QveBEK84IJf6PQ PartitionCount: 1       ReplicationFactor: 1
Configs: min.insync.replicas=1,segment.bytes=1073741824
        Topic: quickstart-events        Partition: 0    Leader: 1       Replicas: 1     Isr: 1  Elr:    Last
KnownElr:
moonwalker@DESKTOP-TOD84J7:~/kafka_2.13-4.1.0$ bin/kafka-console-producer.sh --topic quickstart-events --boo
tstrap-server localhost:9092
This is my first event
>This is my second event
>
```

Messages Verified by Kafka Consumers

```
Windows PowerShell   Ubuntu   Ubuntu   Ubuntu
moonwalker@DESKTOP-TOD84J7:~$ cd kafka_2.13-4.1.0
moonwalker@DESKTOP-TOD84J7:~/kafka_2.13-4.1.0$  bin/kafka-console-consumer.sh
om-beginning --bootstrap-server localhost:9092
Thi is my first event
This is my second event
hello
MoonWalker
^CProcessed a total of 4 messages
moonwalker@DESKTOP-TOD84J7:~/kafka_2.13-4.1.0$
```

Challenges

- No major issues just trouble shooting with Kafka and downloading WSL2 with Ubuntu to make set up easier

Questions

- No major issues just trouble shooting with Kafka and downloading WSL2 with Ubuntu to make set up easier

Action Plan

- **Further environment setup**: Work towards complete integration of Unity, Omniverse, and Kafka
- **Documenting Workflow**: Archive steps, dependencies, and challenges for more efficient set ups, information exchange
- **Continue Research**: Proceed with our research with EC2 and emerging standards regarding Digital Twin Interoperability
- **Validate interoperability**: Confirm omniverse can send data to the API and process responses (or messages delivered via Kafka, if previous step is successful).
    - If Unity setup w/o major issues, will test it in a similar manner to Omniverse